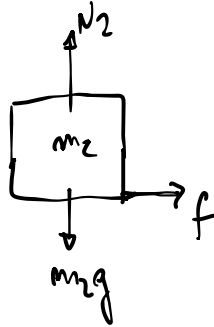
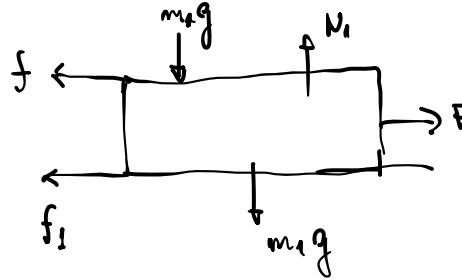


$$I) a_{R1/2} = 0$$

$$II) a_{R1/2} > 0$$



$$\begin{cases} N_2 - m_2g = 0 \\ f = m_2a_2 \leftarrow \end{cases}$$



$$\begin{cases} N_1 - m_1g - m_2g = 0 \\ F - f_1 - f = m_1a_1 \leftarrow \end{cases}$$

$$(I) \quad a_{R1/2} = a_1 - a_2 = 0 \Rightarrow a_1 = a_2 = a$$

$$f \geq \mu_e m_2 g \Rightarrow m_2 a_2 = \mu_e m_2 g \Rightarrow \boxed{a_2 = \mu_e g = a_{max}}$$

$$f_1 = \mu_1 N_1 = \mu_1 (m_1 + m_2) g$$

$$F_{max} - \mu_1 (m_1 + m_2) g - \mu_e m_2 g = m_1 a_{max}$$

$$F_{max} - \mu_1 (m_1 + m_2) g - \mu_e m_2 g = \mu_e m_1 g$$

$$F_{max} = \mu_e (m_1 + m_2) g + \mu_1 (m_1 + m_2) g$$

$$\boxed{F_{max} = (\mu_e + \mu_1) (m_1 + m_2) g} \rightarrow (m_1 + m_2) g = \frac{F_{max}}{\mu_1 + \mu_e}$$

$$F_{max} = \mu M g \quad \begin{cases} \mu = \mu_e + \mu_1 \\ M = m_1 + m_2 \end{cases}$$

$$(II) \quad a_{R1/2} \neq 0 = a_1 - a_2$$

$$\begin{cases} m_1 a_1 = F - \mu_1 (m_1 + m_2) g - \mu_c m_2 g \\ m_2 a_2 = f = \mu_c m_2 g \end{cases}$$

$$a_{R1/2} = a_1 - a_2 = \frac{F}{m_1} - \frac{\mu_1}{m_1} (m_1 + m_2) g - \frac{\mu_c m_2 g}{m_1} - \mu_c g$$

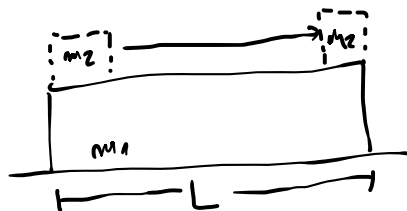
$$a_{R1/2} = \frac{F}{m_1} - \frac{\mu_1}{m_1} (m_1 + m_2) g - \left(\frac{m_2 + m_1}{m_1} \right) \mu_c g$$

$$a_{R1/2} = \frac{F}{m_1} - \left(\frac{m_1 + m_2}{m_1} \right) g (\mu_1 + \mu_c)$$

$$a_{R1/2} = \frac{1}{m_1} \left[F - (\mu_1 + \mu_c) (m_1 + m_2) g \right]$$

$$a_{R1/2} = \frac{1}{m_1} \left[F - \frac{\mu_1 + \mu_c}{\mu_1 + \mu_c} F_{\max} \right]$$

$$F > \frac{\mu_1 + \mu_c}{\mu_1 + \mu_c} F_{\max}$$



$$L = a_{R1/2} \frac{t^2}{2} \Rightarrow$$

$$t = \sqrt{\frac{2L}{a_{R1/2}}}$$